Does Female Leadership Relate to Productivity–Wage Gaps in Colombia?

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Abstract

In labor markets with frictions such as employer concentration, search costs, and limited worker mobility, firms can exercise monopsony power by setting wages below workers' marginal products, thereby generating productivity-wage gaps. The extent of these gaps depends not only on market structure but also on firm managerial structure, which shapes wage-setting and employment policies. A growing literature suggests that female leaders place greater weight on worker welfare and are less likely to engage in anticompetitive behavior. This paper examines whether these tendencies translate into reduced monopsony exploitation, specifically whether female leadership (measured as the share of female executives) attenuates the productivity-wage gap. Using firm-level panel data from the Colombian Annual Manufacturing Survey 2000-2019, I estimate firm-specific productivity-wage gaps following Yeh et al. (2022) and relate them to the female share of firm executives in a linear fixed-effects framework controlling for time-varying firm characteristics, geography, and industry-level factors. I find that increases in female leadership share are associated with reductions in the productivity-wage gap. These findings carry implications for both equity and efficiency: since wages below marginal product depress employment and distort resource allocation, expanding female leadership in emerging markets may therefore yield meaningful labor market gains.

JEL Classification: D43, D63, J16,J42,J7, M12, M52

Keywords: female leadership, labor market power, productivity-wage gap, labor monopsony, Emerging markets, Colombia

1 Introduction

Across the world, workers are paid substantially less than the value they produce. In Colombia, Amodio and de Roux (2024) estimates that workers produce approximately 40 percent more than they earn, a gap that represents not only foregone income for millions of workers, but also a significant source of allocative inefficiency for the broader economy. Understanding why this gap exists, and what can be done to reduce it, is a central question in labor economics.

Standard microeconomic theory predicts that in competitive labor markets, wages equal workers' marginal revenue product. Yet a large and growing body of evidence shows that firms are not wage takers: labor markets are characterized by frictions — including search costs, limited job mobility, and employer concentration — that grant firms wage-setting power.[1] When firms exploit this power, they set wages below workers' marginal productivity, generating a productivity-wage gap commonly referred to as a labor markdown. The consequences extend beyond wages: firms with monopsony power also hire fewer workers than is socially optimal, depressing employment and distorting resource allocation across the economy.

Although monopsony power arises from structural features of labor markets, firms differ substantially in how aggressively they exploit it. This variation suggests that firm-level characteristics beyond market structure shape the productivity-wage gap. One such characteristic is firm ownership. A growing literature documents that female leaders tend to place greater weight on worker welfare and employment stability (Matsa & Miller, 2013; Tunyi et al., 2023), interpret productivity signals differently (Chen et al., 2019; Flabbi et al., 2017), and influence corporate decision-making in ways that may reduce productivity-wage distortions (Flabbi et al., 2019; Levi et al., 2014). At the same time, labor market power has been shown to affect women more strongly than men (Sharma, 2023), raising the possibility that female leadership may also attenuate gender-specific productivity-wage distortions within firms. Together, these strands of evidence motivate the central question of this paper: does greater female leadership reduce the extent to which firms exploit monopsony power?

To examine this question, I use firm-level panel data from the Colombian Annual Manufacturing Survey (EAM, from its Spanish initials), collected by Colombia's National Department of Statistics (Departamento Administrativo Nacional de Estadística (DANE), n.d.) and covering the period 2000–2019. Colombia provides a particularly informative setting for this

[1] A growing literature shows that labor markets are often characterized by frictions. See for instance on search costs (Card, 2022; Azar et al., 2019; Lentz and Mortensen, 2010; Moscarini and Postel-Vinay, 2010; Manning, 2021), on limited job mobility (Caldwell and Danieli, 2024; Schubert et al., 2025; Starr et al., 2021; McAdams, 2019), and on employer concentration (Azar et al., 2017; Benmelech et al., 2018; Rinz, 2022).

analysis. Its manufacturing sector exhibits substantial variation in both female executive representation and labor market concentration, and labor market frictions are more pronounced than in high-income economies, making deviations from competitive wage-setting both more likely and more consequential. I estimate firm-specific productivity-wage gaps following the production-function methodology of Yeh et al. (2022), which recovers firm-level labor markdowns from output and input data without requiring direct information on market structure. I then relate these estimates to the female share of firm executive positions in a linear fixed-effects framework that controls for time-varying firm characteristics and industry-level factors, isolating the within-firm relationship between management composition and wage-setting behavior.

The results reveal a robust negative association between female leadership share and the productivity-wage gap: firms with more women in executive positions set wages closer to workers' marginal product, consistent with less aggressive exploitation of monopsony power. This finding holds across a range of specifications and is not driven by differences in firm size, productivity, or industry composition. These results contribute to two distinct literatures. First, they add to the growing body of work on the determinants of labor market power by showing that management structure shapes the productivity-wage gap. Second, they extend the literature on female leadership and firm behavior by connecting gender composition in executive positions to a concrete labor market outcome with direct welfare implications.

These findings carry important policy implications, particularly for emerging markets where labor market frictions are often more pronounced and monopsony rents are larger. Policies that expand female participation in leader roles through access to finance, legal protections, or targeted investment incentives may generate returns that extend beyond the firms themselves, improving wages and employment for workers more broadly. More generally, the results suggest that the distributional consequences of labor market power are not determined by market structure alone, and that who owns firms matters for how those firms treat their workers.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature and estimations on the productivity-wage gap. Section 3 presents a theoretical framework that illustrates how monopsony power generates a productivity-wage gap and how this gap can be estimated using firm-level data. Section 4 describes the data and presents summary statistics. Section 5 outlines the empirical strategy. Section 6 presents the main results and robustness checks. Finally, section 7 provides the conclusion, policy implications, and directions for future research.

2 Literature Review

Table 1: Summary of Related Literature

Paper	Topic	Data	Main Finding
De Loecker, Eeckhout, & Unger (2020)	Labor market concentration	U.S. manufacturing	Production approach for a robust estimation of elasticities. Findings suggest markups rise not just to cover higher fixed costs, but also due to firms' market power.
Yeh, Macaluso & Hershbein (2022)	Labor markdown estimation	U.S. manufacturing	Most manufacturing plants operate in a monopsonistic environment. Average markdown 1.53 workers earning only 65 cents on the marginal dollar generated.
Amodio & de Roux (2024)	Colombia Labor market power	Colombian manufacturing plants (EAM)	Estimate a firm-level labor supply elasticity of around 2.5, implying that workers produce about 40% more than their wage level
Inter-American Development Bank (2025)	Latin-America and the Caribbean Market Power	Firm-Data LAC (For Colombia EAM)	On average, workers in Latin America and the Caribbean take home in the form of wages 50 percent of what they contribute to their employers.
Matsa & Miller (2013)	Female leadership	Nordic companies (Norway's 2006 quota)	Female leaders tend to emphasize employment stability and worker retention.
Flabbi et al. (2019)	Female executives	Italy employer-employee panel data	Female leadership influences wage-setting and reduce gender gaps within firms.
Levi, Li & Zhang (2014)	Female executives	Aquisition bids by S&P 1500 companies	Female directors less overestimate merger gains and help create shareholder value

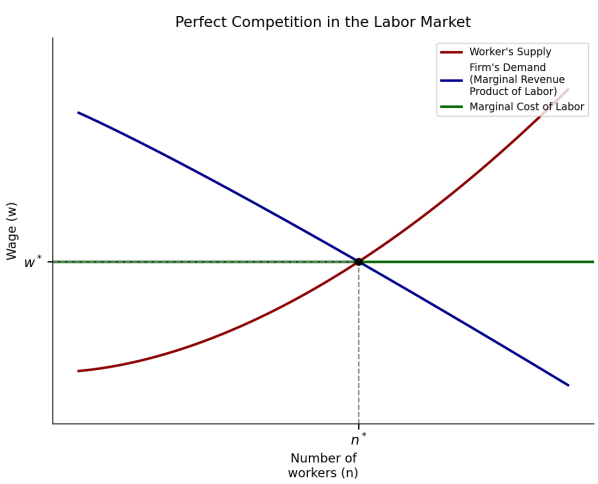
3 Theoretical Framework

To understand the gap between wages and workers' marginal productivity, a useful starting point is the benchmark case of a competitive labor market. As shown in Panel (a) of Figure 1, under perfect competition the marginal cost of labor is constant and equal to the market wage, which each firm takes as given. Firms therefore hire labor up to the point where the marginal revenue product of labor (MRPL) equals the wage.

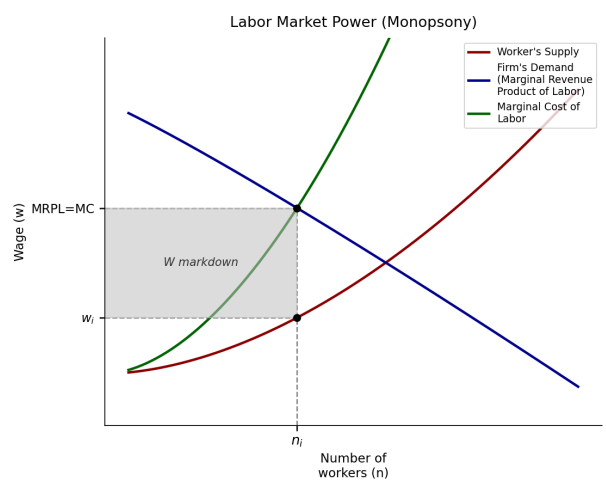
Monopsony power can be exercised by any employer who faces an upward-sloping labor supply curve, as shown in Panel (b) of Figure 1. Hiring an additional worker requires raising the wage not only for the marginal worker but, when wages are set uniformly, for all inframarginal workers as well. As a result, the marginal cost of labor lies strictly above the labor supply curve. The profit-maximizing firm sets employment where MRPL = MCL, yielding both lower employment and a lower wage relative to the competitive benchmark. Crucially, at the optimum, workers are paid a wage w_i determined by the labor supply curve, yet their marginal contribution to firm revenue satisfies MRPL $> w_i$. The wedge between the marginal revenue product of labor and the wage is known as the *markdown*, illustrated by the shaded region in Panel (b), and it is the central object of interest in this paper.

Figure 1: Labor Markets

(a) Perfect Competition



(b) Monopsony



To see how this markdown arises formally, following Ashenfelter et al. (2010) consider a firm's profit maximization problem with labor as the only input. Under perfect competition, the firm takes the market wage $\bar{w}$ as given and chooses labor input L to maximize profits:

$$\max_L \Pi = R(L) - \bar{w}L, \tag{1}$$

Where $R(L) \equiv P * Q(L)$ is the firm's revenue as a function of labor input. The first-order condition for profit maximization equates the marginal revenue with the marginal cost such that:

$$R'(L) = \bar{w} \tag{2}$$

Under monopsony, the firm faces an upward-sloping labor supply curve, so the wage is an increasing function of employment $w(L)$, as shown in figure 1. The profit maximization problem becomes:

$$\max_L \Pi = R(L) - w(L)L, \tag{3}$$

with first-order condition:

$$R'(L) = w(L) + w'(L)L \tag{4}$$

Note that the monopolistic firms also maximizes profits by equating marginal revenue and marginal cost, but the marginal cost of labor is greater than the wage in this case. The right-hand side of the equation exceeds the wage $w(L)$ by the term $w'(L)L$ which captures the additional cost of raising wages for all inframarginal workers when hiring one more.

Rearranging yields to the Lerner index formula for the labor market:

$$\frac{R'(L) - w(L)}{w(L)} = \frac{w'(L)L}{w(L)} \equiv \varepsilon_L^{-1} \tag{5}$$

Where $\varepsilon_L^{-1} \equiv \frac{w'(L)L}{w(L)}$ is the firm's perceived inverse elasticity of labor supply, capturing how much the wage must increase to attract an additional worker. This expression shows that the percentage gap between the marginal revenue product of labor and the wage equals the inverse of the firm's perceived labor supply elasticity: the more inelastic the labor supply (lower ε_L), the larger the gap between the marginal revenue product of labor and the wage.

Following Yeh et al. (2022), I define the labor markdown ν as the ratio of the marginal revenue product of labor to the wage:

$$\nu \equiv \frac{R'(L)}{w(L)} = \varepsilon_L^{-1} + 1 \tag{6}$$

Under this definition, $\nu = 1$ under perfect competition (infinite labor supply elasticity)

and $\nu > 1$ whenever labor supply is upward-sloping. It is therefore sufficient to estimate the firm's perceived labor supply elasticity ε_L to recover the markdown ν. However, as highlighted by Yeh et al. (2022), estimating ε_L in a general setting is challenging, as it requires disentangling the effects of labor market power from those of output market power (monopoly) and from other confounding factors. To address this challenge, I adopt the production approach of De Loecker et al. (2020), which allows for the estimation of firm-level markdowns using data on output and input costs without requiring direct information on market structure.

3.1 Markdown calculation - Production Approach

To understand the production approach from De Loecker et al. (2020) let's consider a simple Cobb-Douglas production function with three inputs: labor L, capital K, and materials M, where L and M adjust frictionlessly within a period, while capital is subject to adjustment frictions. I removed the subscript it for simplicity, but all variables are indexed by time and firm.

The approach relies on the assumption that materials (M) is a fully flexible input satisfying three conditions: it incurs no adjustment costs within a period; its price is taken as given by the firm; and it is chosen statically, in the same period in which output is produced.

The firm's cost minimization problem is:

$$\mathcal{L} = rK + w(L)L + p^m M - \lambda \left(P\bar{Q} - AK^{\theta^K} L^{\theta^L} M^{\theta^M} \right) \tag{7}$$

where θ^K, θ^L, θ^M are the output elasticities of capital, labor, and materials, respectively and A denotes firm-specific productivity. The price of capital is r, the wage function is $w(L)$, and the price of materials is p^m. P is the price of output, $\bar{Q}$ is the quantity of output produced, and λ is the Lagrange multiplier associated with the production constraint which captures the marginal cost of production

The first-order conditions for M and L are:

$$\frac{\partial \mathcal{L}}{\partial M} = p^m - \lambda \theta^M A K^{\theta^K} L^{\theta^L} M^{\theta^M - 1} = 0. \tag{8}$$

$$\frac{\partial \mathcal{L}}{\partial L} = w'(L)L + w(L) - \lambda \theta^L A K^{\theta^K} L^{\theta^L - 1} M^{\theta^M} = 0, \tag{9}$$

We can simplify the first-order conditions as:

$$p^m = \lambda \theta^M \frac{Q}{M} \tag{10}$$

$$w'(L)L + w(L) = \lambda\theta^L \frac{Q}{L} \tag{11}$$

The Lagrange multiplier λ captures the marginal cost of production, thus, the firm's markup, defined as the price-to-marginal-cost ratio is therefore:

$$\mu = \frac{P}{\lambda}. \tag{12}$$

Substituting and rearranging in the first-order conditions for materials:

$$p^m = \theta^M \frac{PQ}{M\mu}, \tag{13}$$

We obtain an expression for the markup in terms of observable quantities:

$$\mu = \theta^M \frac{PQ}{p^m M}, \tag{14}$$

For the first order condition for labor, let's first express the left-hand side in terms of the labor supply elasticity $w'(L)L + w(L) \equiv \frac{w'(L)L}{w(L)} + 1 \equiv (\varepsilon_L^{-1} + 1)w(L)$. Rearranging the first-order condition for labor gives:

$$(\varepsilon_L^{-1} + 1)w(L) = \frac{\theta^L}{\mu}\frac{PQ}{L}, \tag{15}$$

Now, we know from the theory section that the labor markdown ν is defined as the ratio of the marginal revenue product of labor to the wage, which can be expressed as:

$$\nu \equiv \frac{\text{MRPL}}{w(L)} = \varepsilon_L^{-1} + 1. \tag{16}$$

Substituting $\varepsilon_L^{-1} + 1 = \nu$ into the first-order condition for labor gives:

$$\nu = \frac{\theta^L}{\mu}\frac{PQ}{Lw(L)}. \tag{17}$$

Substituting the expression for the markup μ derived from the first-order condition for materials into the above equation for markdown ν yields:

$$\nu = \frac{\theta^L}{\theta^M} \cdot \frac{p^m M}{w(L)L}. \tag{18}$$

This expression shows that the markdown (productivity-wage gap) can be estimated using the ratio of the firm's expenditure on materials to its wage bill, scaled by the ratio of output elasticities. The output elasticities θ^L and θ^M are estimated from the production

function, while the input expenditures are computed from observed data on wages, materials, output, and prices. This approach allows for the estimation of firm-level markdowns while controlling for both output and labor market power.

4 Data

This paper uses firm-level data from the Annual Manufacturing Survey (*Encuesta Anual Manufacturera*, EAM) conducted by Colombia's National Administrative Department of Statistics (DANE). The EAM is a census of medium and large manufacturing firms with more than 10 employees that provides detailed and standardized information on firms' production, input use, employment composition, and financial accounts. I restrict the analysis to the period 2000–2019 to ensure data consistency and to avoid disruptions associated with the COVID-19 pandemic and exclude the information before 2000 as the data in that period had limited information on executive positions by gender. Although the data are collected at the establishment level, I aggregate to the firm level by summing across all plants belonging to the same firm in a given year using the unique firm identifier provided in the survey, thereby capturing the full scope of each firm's operations as some managers might operate only from the headquarters office of the firm but still have impact on the plants.

Administrative firm-level data frequently contain extreme values arising from measurement error, recording mistakes, or genuine but highly influential outliers, which can bias production-function estimates and, by extension, the labor market power calculations that follow. I apply a series of sample restrictions to address this concern.

Following Yeh et al. (2022), I apply three filters to the key variables: output, labor, capital, and materials. First, I winsorize each variable at the top and bottom 1 percent of its distribution to limit the influence of extreme observations. Second, I drop observations with non-positive values for output or any input, as these are inconsistent with the production framework. Third, I compute each firm's materials cost share,defined as materials expenditure as a share of total costs, and exclude observations falling below one quarter of the sector-year median. Firms with anomalously low materials shares are unlikely to satisfy the conditions required for identification: no adjustment costs, static input choice, and the absence of input market power. Their inclusion could therefore contaminate the proxy-variable estimator.

Beyond these filters, I drop observations in which the total number of owners or directors is zero, the total number of workers is zero. Taken together,these restrictions exclude observations representing 7.99 percent of total gross output in the raw sample, making the final dataset still representative of the manufacturing sector in Colombia. The final dataset

is an unbalanced panel of 11,235 Colombian manufacturing firms observed over 2000–2019, yielding 53,688 firm-year observations spanning 105 industries and 23 geographic regions.

Table 2 reports descriptive statistics for the main variables used in the empirical analysis. The central variable of interest is the share of female executives, defined as the proportion of women among the firm's owners and senior managers. The sample mean stands at 45.2 percent, with a median of 49.8 percent, suggesting that the distribution is approximately symmetric. Nonetheless, the standard deviation of 39.9 percentage points—alongside the full support of the distribution from 0 to 100 percent—indicates considerable cross-sectional heterogeneity: a non-trivial share of firms are either entirely male-led or entirely female-led.

Firms in the sample are, on average, small. The mean (median) number of workers is 56 (20), with a standard deviation of 146 and a maximum of 4,921 employees, revealing a right-skewed size distribution consistent with the fact that manufacturing sectors in developing economies are dominated by small and medium-sized enterprises Alviarez et al. (2025). The firm's share of industry employment averages 0.02 percent, with a median of 0.004 percent, confirming that most firms are tiny iin terms of employment relative to their respective industries. The large standard deviation (0.066 percentage points) and maximum value (1 percent) indicate, however, that a small number of firms command disproportionate shares of sectoral employment—a pattern consistent with the firm concentration patterns documented by Alviarez et al. (2025). As markdowns are expected to be larger in more concentrated markets, this variable is crucial for the empirical analisys to isolate the effect of the labor market structure.

On average, 62.0 percent of the workforce is engaged in production activities, with a median of 66.7 percent and a standard deviation of 21.7 percentage points. The range spanning from 0 to 100 percent reflects meaningful variation in the degree to which firms are production- versus administration-intensive, which may proxy for differences in technology, capital intensity, or organizational complexity across the sample.

Mean gross output—proxied by total sales at factory prices net of direct taxes—is $1.22 million USD, but the distribution is heavily right-skewed: the median of $11.36 thousand USD lies far below the mean, and the standard deviation of $536.01 million USD is orders of magnitude larger than the central tendency, pointing to the presence of a small number of very large firms that dominate the upper tail. This pattern is a well-established feature of firm-level data and underscores the importance of controlling for firm size in all empirical specifications as Yeh et al. (2022) found that this variable is significant for the estimation of markdowns. The average wage per worker is $1.83 thousand USD annually, with a median of $2.30 thousand USD. The dispersion in wages—ranging from $0.35 to $1,498.01 thousand USD—likely reflects both industry-level wage premia and substantial differences in skill

composition and capital-labor ratios across firms.

The average firm in the sample is approximately 4.3 years old, with a median of 3 years and a maximum of 20 years. The right skew in the age distribution, combined with a standard deviation of 3.4 years, suggests that the sample is populated predominantly by young and recently established firms. This is consistent with high rates of entry and exit that are characteristic of manufacturing sectors in emerging market economies Alviarez et al. (2025), and motivates the inclusion of firm age as a control variable to account for lifecycle effects even though Yeh et al. (2022) found that it is not significant for the U.S. manufacturing sector.

Taken together, the descriptive statistics paint a picture of a sample dominated by small, young, production-intensive firms exhibiting wide variation in size, wages, output, and leadership composition. The pronounced dispersion across all dimensions underscores the importance of the empirical framework developed in the following sections, which is designed to exploit within-firm variation over time while controlling for the rich observed heterogeneity documented here.

Table 2: Descriptive Statistics

Variable	Mean	SD	Min	Median	Max
Share of female executives (%)	45.192	39.925	0.000	49.812	100.000
Firm's share of industry employment (%)	0.020	0.066	0.000	0.004	1.000
Share of workers in production (%)	61.951	21.743	0.000	66.667	100.000
Gross output ($ millions USD)	1.219	536.010	756.959	11.356	2,968.214
Firm age (years)	4.262	3.422	1	3	20
Number of workers	56	146	1	20	4921
Wage per worker ($ thousands USD)	1.827	91	349.180	2.296	1,498.014

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: The wage variable corresponds to the total wage bill of the firm divided by the number of employees, while gross output is proxied by total sales at factory price without direct taxes. To simplify the presentation, all dollar values are calculated using an exchange rate of 1 USD = 3,670 COP.

4.1 Markdown Estimation

The main outcome variable is the firm-level labor markdown, which captures the gap between a worker's marginal revenue product and their wage. Following Yeh et al. (2022), the markdown can be expressed in terms of observable quantities as:

$$\nu_{it} = \frac{\theta_{it}^{L}}{\theta_{it}^{M}} \cdot \frac{p_{it}^{m} M_{it}}{w_{it} L_{it}}, \tag{19}$$

where $p_{it}^{m} M_{it}$ is total expenditure on intermediate inputs and $w_{it} L_{it}$ is the total wage bill,

both observed directly in the data. To recover firm-year output elasticities θ_{it}^L and θ_{it}^M, I estimate a translog gross-output production function following De Loecker et al. (2020):

The values $p^m M$ and $w(L)L$ can be extracted directly from the dataset, $p^m M$ is the total expenditure on raw materials, and $w(L)L$ is the total wage bill which corresponds to the total remuneration paid to permanent and temporary employees, both directly hired and outsourced. To estimate θ^L and θ^M, I use De Loecker et al. (2020) code which uses a translog production function with polynomial terms as follows:

$$\begin{aligned}\log(GO_{it}) = \beta_0 &+ \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + \beta_3 \log(M_{it}) \\ &+ \beta_4 (\log K_{it})^2 + \beta_5 (\log L_{it})^2 + \beta_6 (\log M_{it})^2 \\ &+ \beta_7 \log(K_{it}) \log(L_{it}) + \beta_8 \log(K_{it}) \log(M_{it}) + \beta_9 \log(L_{it}) \log(M_{it}) \\ &+ \gamma_t + \varepsilon_{it}.\end{aligned} \tag{20}$$

where GO_{it} is gross output, K_{it} is capital, L_{it} is labor, M_{it} is intermediate inputs, γ_t are year fixed effects, and ε_{it} is the error term. Labor is measured as the total number of employees. Capital is proxied by the book value of fixed assets, with its rental cost computed via a user-cost formula that incorporates asset-specific depreciation rates from De Loecker et al. (2020) and year-specific interest rates and inflation from the Central Bank of Colombia (Banco de la República de Colombia, n.d.). Intermediate inputs are measured as total expenditure on raw materials, intermediate goods, and packaging; consistent with the requirements of the proxy-variable estimator, materials serve as the flexible input in production. Gross output is proxied by firm-level total sales at factory prices, net of direct taxes, as reported in the Colombian Manufacturing Survey (EAM).

The production function is estimated separately for each 4-digit ISIC sector via the two-stage GMM procedure of De Loecker et al. (2020), which corrects for the endogeneity of input choices.[2] Given the translog structure, output elasticities are firm-year specific and recovered as the partial derivatives of output with respect to each input:

[2]Because firms observe their own productivity before choosing inputs, OLS estimates are biased upward. The estimator recovers a productivity proxy $\hat{\phi}_{it}$ from a first-stage OLS regression, then constructs the productivity shock $\omega_{it}(\beta) = \hat{\phi}_{it} - \mathbf{x}'_{it}\beta$ for a candidate parameter vector β. Assuming a Markov process for productivity, its conditional expectation is approximated by a cubic polynomial in ω_{it-1}. The innovation $\xi_{it} = \omega_{it} - E[\omega_{it} \mid \omega_{it-1}]$ is orthogonal to lagged inputs by assumption, providing the GMM moment conditions. Capital is treated as predetermined and enters in current values; labor and materials are instrumented with their lagged values and lagged polynomial terms. The GMM criterion is minimized using a Nelder-Mead optimizer, with a gradient-based Rvmmin fallback when convergence is not achieved.

$$
\begin{aligned}
\hat{\theta}_{K,it} &= \hat{\beta}_1 + 2\hat{\beta}_4 \log(K_{it}) + \hat{\beta}_7 \log(L_{it}) + \hat{\beta}_8 \log(M_{it}), \\
\hat{\theta}_{L,it} &= \hat{\beta}_2 + 2\hat{\beta}_5 \log(L_{it}) + \hat{\beta}_7 \log(K_{it}) + \hat{\beta}_9 \log(M_{it}), \\
\hat{\theta}_{M,it} &= \hat{\beta}_3 + 2\hat{\beta}_6 \log(M_{it}) + \hat{\beta}_8 \log(K_{it}) + \hat{\beta}_9 \log(L_{it}).
\end{aligned} \tag{21}
$$

For sectors where the translog GMM fails to converge or yields non-positive elasticities, I fall back to a Cobb-Douglas specification estimated via the procedure of Ackerberg et al. (2015), which imposes sector-constant elasticities but is considerably more robust to limited panel variation.[3]

Table 3 reports the distribution of estimated output elasticities, markups, and markdowns. The estimated output elasticities are broadly consistent with the existing literature and lend credibility to the markdown estimates. The mean output elasticity of labor is 0.33 and the mean elasticity of materials is 0.63, with capital averaging 0.12 — a pattern consistent with the capital-light, assembly-intensive nature of much Colombian manufacturing and in line with estimates reported by De Loecker et al. (2020) and Yeh et al. (2022) for U.S. manufacturing using the same methodology. The dominance of materials in the cost structure further supports their use as the flexible input in the proxy-variable estimator.

The mean product markup is 1.22, implying that the average firm prices output approximately 22 percent above marginal cost. The median of 1.33 falls within the range estimated by De Loecker et al. (2020) for U.S. manufacturing, and is considerably below the average of 87 percent above marginal cost reported for the broader Latin American and Caribbean region by Inter-American Development Bank (2025).

Turning to the markdown estimates, the median firm-level markdown is 1.68,implying that the median worker generates roughly 68 percent more revenue at the margin than they take home in wages — equivalently, for every dollar generated at the margin, the worker receives approximately 60 cents. This exceeds the median of 1.53 estimated for U.S. manufacturing by Yeh et al. (2022), consistent with stronger monopsony power in the Colombian labor market. The estimate is also broadly aligned with Amodio and de Roux (2024), who report a labor supply elasticity of approximately 2.5 for Colombia, implying workers produce around 40 percent more than their wage. The substantial dispersion in markdowns — the interquartile range runs from 0.58 to 3.24 — mirrors the firm-level heterogeneity documented in the literature (Yeh et al., 2022) and motivates the regression analysis in Section 5.

[3]The Cobb-Douglas specification takes the form $\log(GO)_{it} = \beta_0 + \beta_K \log(K)_{it} + \beta_L \log(L)_{it} + \beta_M \log(M)_{it} + \gamma_t + \varepsilon_{it}$, estimated with the same two-stage GMM logic. Because elasticities are sector-constant under Cobb-Douglas, I impute the sector-level estimates to all firms in that sector. The vast majority of firm-year observations are covered by the translog estimator.

Table 3: Production Function Parameters

Variable	*Mean*	*p25*	*Median*	*p75*
$\hat{\theta}_{L,it}$ Output elasticity of labor	0.326	0.168	0.304	0.381
$\hat{\theta}_{K,it}$ Output elasticity of capital	0.117	0.066	0.101	0.143
$\hat{\theta}_{M,it}$ Output elasticity of materials	0.631	0.553	0.625	0.723
$\hat{\epsilon}_{it}$ Total factor productivity	0.573	2.333	3.285	3.968
$\hat{\mu}_{it}$ Markup	1.215	1.025	1.330	1.800
$\hat{\nu}_{it}$ **Markdown (productivity-wage gap)**	**0.835**	**0.576**	**1.680**	**3.242**

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: Output elasticities are estimated via the translog production function approach of De Loecker et al., 2020, with a Cobb-Douglas fallback following Ackerberg et al., 2015 for sectors where the translog GMM does not converge. Markups and markdowns follow Yeh et al., 2022.

5 Empirical Framework

To examine the relationship between labor market power—as measured by the firm-level productivity-wage gap— and female leadership, I estimate the following multiple linear regression model:

$$\log(\text{Markdown}_{it}) = \alpha + \beta\,\text{FemaleLeadership}_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it}, \tag{22}$$

The dependent variable $\log(\text{Markdown}_{it})$ is the logarithm of the productivity-wage gap for firm i in year t. The markdown is defined as the ratio of the firm's marginal revenue product of labor to its wage bill. The logarithmic transformation is applied to address positive skewness in the raw markdown distribution. $\text{FemaleLeadership}_{it}$ denotes the share of women in executive and ownership positions at firm i in year t. The coefficient of interest, β, has a semi-elasticity interpretation, and it captures the percentage change in markdown to a one percentage point increase in the share of female executives, holding other factors constant.

The vector $\mathbf{X}_{it}$ includes a rich set of firm-level controls designed to isolate the leadership channel from confounding sources of variation. Firm size is captured through employment deciles, which allow for a flexible, non-parametric relationship between scale and markdown, using gross output, which controls for revenue-side scale effects on wage-setting capacity, the reference group for the estimation is firms at the bottom of the distribution that are in the decile 1. Firm age enters as a categorical variable, grouping firms into early (2-5 years), established (6-10 years), mature (11-15 years), and old (16-20 years), new firms that are 1

year old or less are the reference group. This accounts for lifecycle dynamics in labor market power, since older firms may accumulate monopsony rents through reputation, worker attachment, and reduced job-to-job mobility Yeh et al. (2022). Total factor productivity (TFP), recovered from the production function estimation, is included to absorb the positive correlation between productivity and markdowns, and to address the concern raised by Moreno-Gómez and Calleja-Blanco (2018) that female-led firms in Colombia are systematically more productive. I also include the share of employment by the firm in the industry to control for the firm labor market power, and the industry-level Herfindahl-Hirschman Index (HHI) to control for product market concentration. I also control for the share of production workers to account for differences in the composition of the workforce across firms, which may affect both the markdown and the share of female executives. The inclusion of these controls is crucial to isolate the effect of female leadership on labor market power from confounding factors related to firm size, age, productivity, industry structure, and workforce composition.

The specification saturates the model with three sets of fixed effects. Year fixed effects (α_t) absorb aggregate macroeconomic fluctuations, including business-cycle variation in labor market tightness and economy-wide trends in productivity and wages. Sector fixed effects (α_s) account for permanent cross-industry differences in production technology, the degree of product market competition, and the structural propensity toward labor market concentration. Region fixed effects (α_r) control for persistent local labor market conditions, including geographic variation in workers' outside options, commuting costs, and the density of competing employers. In the most demanding specification, all three sets of fixed effects are included simultaneously, so that identification of β comes from within-sector, within-region, within-year variation in female leadership and markdowns across firms. Standard errors are clustered at the firm level throughout to account for serial correlation in the markdown within firms over time.

6 Results

Taking into account that the data is panel, standard errors are clustered at the firm level to account for serial correlation in the markdown within firms over time.

Table 4: Regression Results - Dependent variable: log(Markdown)

	(1)	(2)	(3)	(4)
Female Leadership	-0.0010***	-0.0010***	-0.0011***	-0.0003 .
	(0.0001)	(0.0002)	(0.0002)	(0.0002)
Labour share (industry)		2.0356***	2.4268***	-1.0845***
		(0.3723)	(0.3713)	(0.3198)
Share of production workers		-0.0021***	-0.0018***	-0.0013***
		(0.0003)	(0.0003)	(0.0003)
Industry HHI		0.5729***	0.5719***	0.0913 .
		(0.0634)	(0.0635)	(0.0547)
Firm decile 2		0.0361	0.0510.	0.0104
		(0.0290)	(0.0288)	(0.0223)
Firm decile 3		0.0171	0.0328	-0.0422 .
		(0.0288)	(0.0286)	(0.0222)
Firm decile 4		-0.0428	-0.0159	-0.1071***
		(0.0291)	(0.0290)	(0.0225)
Firm decile 5		0.0733*	0.1043***	-0.0818***
		(0.0298)	(0.0296)	(0.0231)
Firm decile 6		0.1508***	0.1853***	-0.0239
		(0.0304)	(0.0303)	(0.0237)
Firm decile 7		0.1968***	0.2385***	0.0170
		(0.0313)	(0.0312)	(0.0245)
Firm decile 8		0.3773***	0.4155***	0.1575***
		(0.0325)	(0.0324)	(0.0256)
Firm decile 9		0.4822***	0.5162***	0.2371***
		(0.0336)	(0.0336)	(0.0266)
Firm decile 10		0.6640***	0.6782***	0.4416***
		(0.0391)	(0.0390)	(0.0315)
Age: Early (2–5 yrs)		-0.0383	-0.0554*	-0.0456*
		(0.0267)	(0.0265)	(0.0205)
Age: Established (6–10 yrs)		-0.0170	-0.0550.	-0.0143
		(0.0294)	(0.0293)	(0.0228)
Age: Mature (11–15 yrs)		0.0932*	0.0253	0.0471
		(0.0389)	(0.0389)	(0.0302)
Age: Old (16–20 yrs)		0.1025	0.0091	0.1161*
		(0.0683)	(0.0681)	(0.0528)
TFP		0.1557***	0.1578***	0.0801***
		(0.0039)	(0.0039)	(0.0036)
FE: Year	X	X	X	X
FE: Region	X		X	X
FE: Industry	X			X
N	48,292	31,664	31,664	31,664
R^2	0.5097	0.1243	0.1386	0.4849

Note: $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Across specifications (1) through (3), the estimated coefficient on female leadership is negative and robust to the inclusion of the full set of firm-level controls and region fixed effects. In Column (3) a 10 percentage-point increase in the female executive share is associated with a 1.1 percent reduction in the log markdown ($\hat{\beta} = -0.0011$, $p < 0.001$). To gauge the economic magnitude of this estimate, consider the median firm in the sample, whose markdown of 1.680 implies that workers receive approximately 59.5 cents for every dollar of output they generate. A 10 percentage-point increase in female leadership would compress the markdown to approximately 1.662 ($1.680x \exp(-0.0011 \times 10)$), raising the worker's share to roughly 60.2 cents per dollar produced—a gain of approximately 0.7 cents on the dollar ($7 dollars gain for every $100 dollars produced), holding all other firm characteristics constant.

The coefficient attenuates to -0.0003 and loses conventional significance once industry fixed effects are added, indicating that a meaningful share of the cross-sectional covariation between female leadership and markdowns operates through systematic sorting of female-led firms into industries with structurally lower labor market power.

Firm Size. The employment decile indicators reveal a robust, monotonically increasing relationship between firm size and the markdown. Relative to the smallest decile, firms in the top decile exhibit a log markdown that is 0.44 log points higher in the preferred specification (Column 4), a pattern consistent with the theoretical prediction that larger firms face less elastic labor supply curves and therefore exercise greater monopsony power Ashenfelter et al. (2010). The relationship is statistically significant from the sixth decile onward and accelerates sharply in the upper tail of the size distribution.

TFP and Labor Composition. The TFP coefficient is positive and highly significant across all specifications ($\hat{\gamma}_{\text{TFP}} = 0.080$ in Column 4, $p < 0.001$), confirming that more productive firms pay workers a smaller fraction of their marginal product—consistent with a rent-extraction mechanism in which productivity gains accrue disproportionately to firm owners rather than workers cite someone. The share of production workers enters with a negative and significant coefficient, suggesting that more production-intensive firms—which typically employ lower-skilled workers with fewer outside options—exhibit smaller markdowns, perhaps reflecting compressed wage structures in the lower tail of the skill distribution.

Market Structure. The industry HHI enters positively and significantly in Columns (2) and (3), consistent with the view that product market concentration amplifies labor market power by reducing the number of competing employers cite someone. The coefficient be-

comes marginally significant in the preferred specification once industry fixed effects absorb the time-invariant component of market concentration, confirming that the HHI effect is primarily structural rather than cyclical.

Firm Age. Relative to the youngest cohort (firms aged one year), early-stage firms (2–5 years) exhibit lower markdowns in the preferred specification ($\hat{\gamma} = -0.046$, $p < 0.05$), consistent with the notion that very young firms may initially underprice labor to attract workers before accumulating the reputation and worker attachment that sustain monopsony rents over time cite someone. Coefficients for older age cohorts are positive but imprecisely estimated in most specifications, suggesting diminishing lifecycle effects beyond the early growth phase.

Model Fit. The R^2 rises from 0.124 in Column (2) to 0.485 in the fully saturated Column (4), with the large jump attributable primarily to the inclusion of industry fixed effects. This pattern underscores the importance of controlling for permanent industry-level heterogeneity in production technology and market structure when studying firm-level markdowns.

7 Conclusion

to be done

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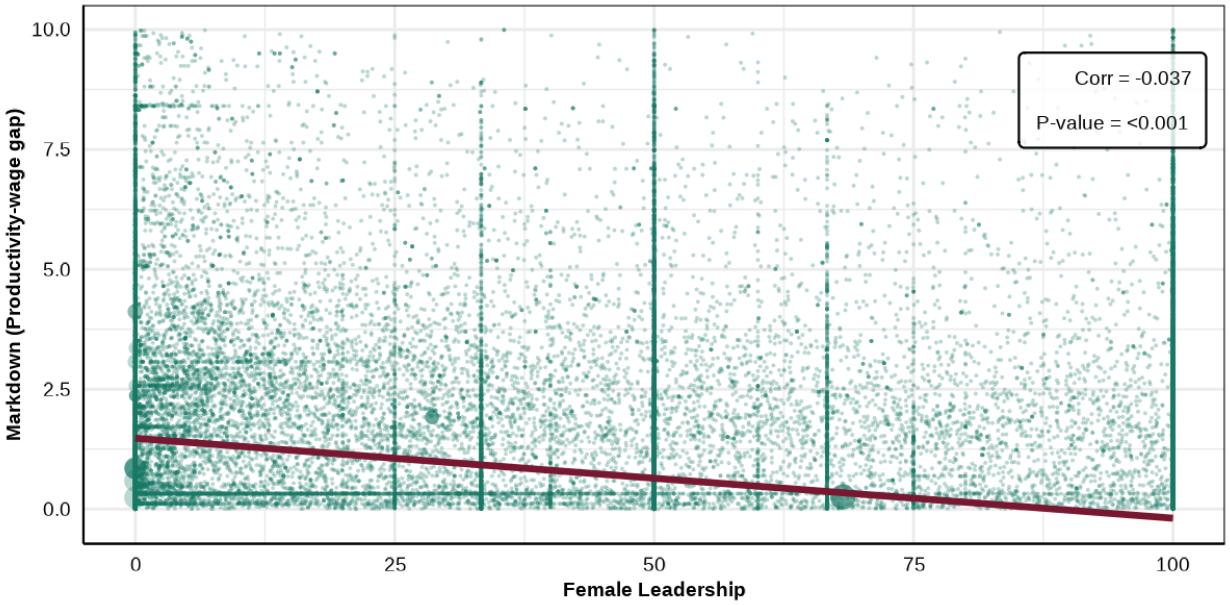
Appendix: Correlation between Markdown and Female Leadership

Figure 2 plots the firm-level estimated markdown against the female share of ownership, with scatter points weighted by each firm's gross output. The correlation is negative ($\rho = -0.037$, $p < 0.001$), offering preliminary evidence that greater female participation in leadership positions is associated with workers capturing a larger share of their marginal revenue product.

For visual clarity, the figure is truncated at markdown values below 10; however, the reported correlation is computed over the full sample of observations.

Figure 2: Markdown and Female Leadership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM), 2000–2019.